# The Dark Universe Under Scrutiny: Null Results, Wavering Dark Energy, and Contested Evidence

## Introduction

Dark matter and dark energy together account for most of the cosmos, yet neither has been directly explained. This report examines three fronts in the effort to understand them. First, LZ and XENONnT's null results are steadily shrinking the WIMP parameter space and approaching the neutrino fog, pushing theory toward axions and other candidates. Second, DESI data hint that dark energy evolves, but the signal's significance depends heavily on supernova, CMB, and tracer choices. Third, the Bullet Cluster, often called a smoking gun for dark matter, proves a model-dependent test that modified gravity has not clearly failed.

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Dark matter and dark energy are inferred from gravitational effects rather than detected directly. Three lines of work show how contested the evidence is. Direct-detection experiments keep returning null results that narrow the allowed parameter space. Cosmological data hint that dark energy may evolve in time, but the significance is fragile. Cluster-collision evidence for dark matter is model-dependent and less settled than popular accounts suggest.

**Direct detection: null results that sharpen the picture.** Dark matter is inferred to make up roughly 85 percent of the universe's mass. Galaxy rotation curves, lensing and the cosmic microwave background all support this, but its particle identity is unknown. Candidates include WIMPs, axions, sterile neutrinos and primordial black holes. Liquid-xenon time projection chambers lead the WIMP search. The LUX-ZEPLIN (LZ) collaboration has analyzed 417 live days of data, the largest dataset any dark matter detector has gathered. It found no WIMPs between 3 and 9 GeV/c², the first time LZ has searched below 9 GeV/c², and it reports world-leading limits above 5 GeV/c² [3][5]. The XENON program shows the same trend. XENON100's limit of 1.1 × 10■■■ cm² at 50 GeV/c² was followed by XENON1T's 4.1 × 10■■■ cm² at 30 GeV/c², a gain of about a factor of 27, with no signal [4].

Two developments complicate the story. LZ detected solar neutrinos for the first time, which shows the detectors are very sensitive. It also means they are approaching the "neutrino fog," where neutrino backgrounds can mimic or mask WIMP signals [3][5]. Coverage frames this as detectors now

testing electroweak physics, with each null result constraining specific models [2]. Anomalies also come and go. XENON1T's electron-recoil excess, potentially consistent with axion dark matter, was ruled out by XENONnT's first results [4]. The combined LZ and XENONnT null results are cited as motivation to probe alternative models with the same data [1]. The field's response is XLZD, a larger detector combining the best of LZ, XENONnT and DARWIN. It would push the WIMP search toward the neutrino fog and also probe dark photons, axion-like particles, neutrinos and cosmic rays [3]. Complementary approaches such as axion haloscopes remain necessary.

**Dark energy: a consistent direction, uncertain significance.** The standard ΛCDM model treats dark energy as a cosmological constant with w = −1. Baryon acoustic oscillation (BAO) measurements from the Dark Energy Spectroscopic Instrument (DESI), combined with CMB and supernova data, persistently prefer time-evolving dark energy in the CPL parameterization [7]. Supernova-only results were mild. Pantheon+ gave w = −0.90 ± 0.14, and Union3 showed a 1.7–2.6σ tension with ΛCDM [6]. DESI's first-year BAO data raised the tension to 2.5σ, 3.5σ and 3.9σ when combined with CMB and Pantheon+, Union3 or DES-SN5YR respectively [6][7]. DESI DR2 with CMB alone excludes ΛCDM at 3.1σ. Adding supernovae gives 2.8σ, 3.8σ and 4.2σ for the same three samples [9].

Across eight dark energy parameterizations, every model lands in the same region of parameter space, w■ > −1 and w■ < 0. This is the "Quintom-B" pattern, in which dark energy is quintessence-like today but crossed the phantom divide in the past, most often near z ~ 0.5 [6][10]. The GEDE model is an exception for the supernova combinations, because its w(z) stays above −1 and never crosses [10]. Several caveats limit the finding:

- Within DESI DR2, the preference is driven mainly by the LRG1–2 tracers, which raises the risk of an underconstrained and unstable inference [9].
- Bayesian evidence is only weak to moderate and depends on the dataset. No model consistently beats ΛCDM, which stays statistically

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## Conclusion

Across all three lines of inquiry, the dark universe proves harder to pin down than its headline claims suggest. Xenon detectors such as LZ and XENONnT have found no WIMPs, but their null results steadily narrow the viable parameter space while approaching the neutrino fog. DESI data point consistently toward evolving dark energy, yet the significance shifts with the supernova sample, CMB probe, and individual BAO tracers, so ΛCDM survives. The Bullet Cluster remains a contested, model-dependent test of

dark matter against modified gravity. Progress now depends on better calibration, independent datasets, and explicit, falsifiable modeling.